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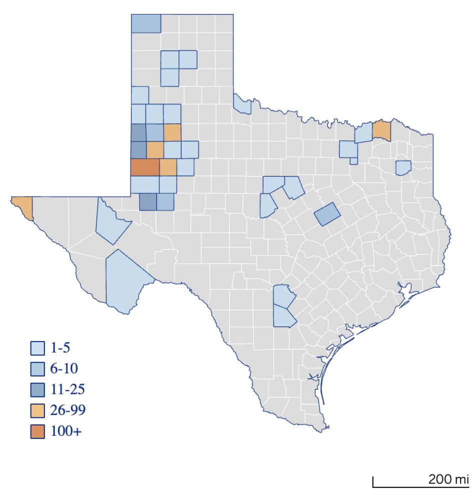
Predictors of Kindergarten Measles Immunization Rates

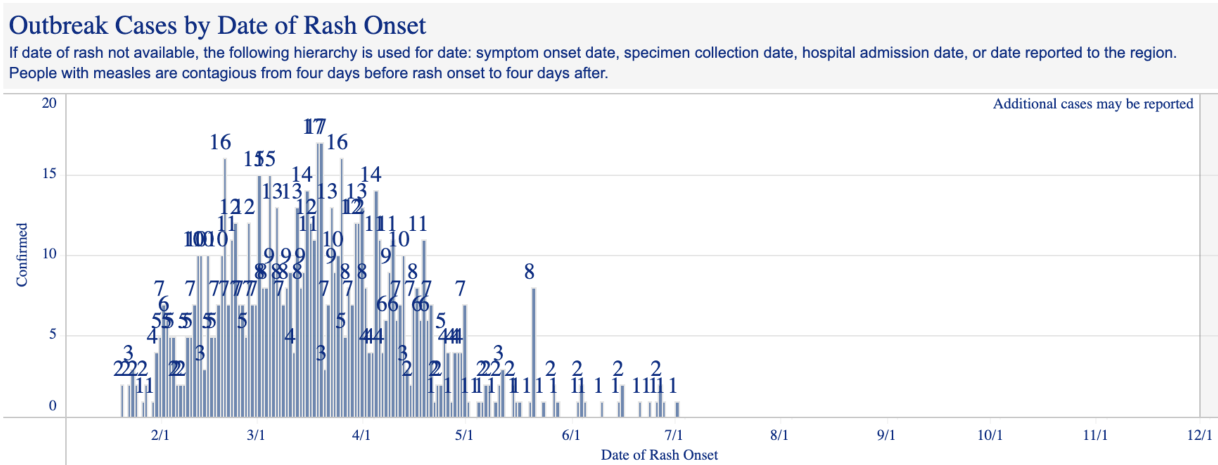
Problem statement

A recent analysis documented a decrease in the county-level proportion of American kindergarteners that are up-to-date with the recommended measles-mumps-rubella (MMR) (Dong et al. 2025) since the COVID-19 pandemic. Analysis of national CDC vaccination coverage and exemption data for public school kindergarteners found small declines between 2019 and 2024 in the coverage of diphtheria, tetanus, and acellular pertussis vaccine (DTaP, from 94.9 to 92.5-0.4%), two doses of MMR (from 95.2% to 92.1%), as well as an increase in the proportion of kindergarten children with an exemption from one or more required vaccinations Williams et al. 2025. Although there are variations across states and counties, these data suggest immunization rates among kindergarten aged children have not recovered to pre-pandemic levels (article) and fall short of the 95% threshold set by the WHO to prevent disease outbreaks (WHO, UNICEF, 2025).

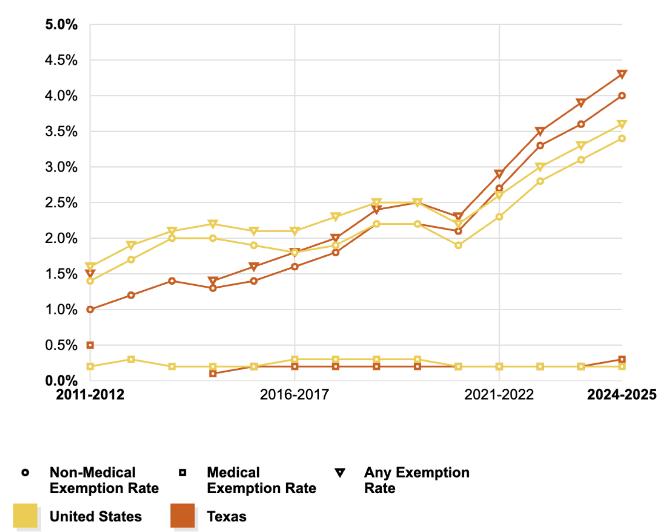


In early 2025, a measles outbreak among children in Western Texas garnered national attention drawing attention to inadequate child immunization rates in American communities. What happened? Between January and July 2025, 762 cases of measles were confirmed, about 29.5% of them in children under the age of five years old. The pace of the outbreak in Texas, concentrated in Gaines County eventually slowed and was declared by the Texas Department of State and Health Services (TX DSHS) to have ended on August 12th. During the West Texas measles outbreak, measles also spread across state borders to neighboring counties, in New Mexico, Kansas, and Iowa (Forbes article, map sourced from TX DSHS website).

Figure 5



In the United States, immunization requirements for children entering school are set by state governments while national entities such as the Advisory Committee on Immunization Practices (ACIP) make recommendations which influence which vaccines insurance companies will pay for children to receive. All states have mandatory vaccinations for children entering kindergarten in public schools and all but three states allow for non-medical exemptions for reasons such as religious beliefs (KFF article, accessed Dec 22, 2025). In June 2025, Texas passed a bill (HB 1586) that made it easier for caregivers to request non-medical exemptions to the state's health department. The same month, another bill (HB 3304) that would have prohibited child care facility and school immunizations mandates in favor of voluntary recommendations, died but this bill did not make it into law. While the rate of exemptions in Texas is only slightly above the national average (4.0% vs. 3.4% nationwide), this metric has been steadily rising in Texas and across the country over the last 15 years. (KFF Vaccination Exemption Rates Among Kindergartners)



The West Texas outbreak was quickly linked to non-vaccination in schools (Fitzpatrick et al). Most of the individuals who contracted measles were unvaccinated (95% of cases) or their vaccination status was unknown (3% of cases, Texas Department of State Health Services, accessed Aug 8, 2025). Given the importance that widespread vaccination plays in protecting vaccinated individuals as well as the unvaccinated or vulnerable persons around them, this project aims to investigate the factors that influence county-level child immunization rates against measles by triangulating available county-level data within the state of Texas.

Primary Questions:

1. What are the strongest predictors of high county-level measles vaccination rates across Texas being up-to-date on the MMR vaccine series?
2. What ability do machine learning models have in predicting future (i.e. 2023) county-level measles vaccination immunization coverage rates based on data from previous years?

Data Wrangling

Data Sources

Several potential data sources for measles-containing vaccine outcomes were considered for this analysis including the Texas Immunization Registry (ImmTrac2), the National Immunization Survey (NIS-Child) for children 0-24 months, and school-vaccination coverage data from the DSHS website. Unfortunately, NIS-Child data is only available to the public at the state-level. ImmTrac2 data was available at the county-level but only provided vaccine dose counts for each county and not the number of children that received one or more doses of measles-preventing vaccine. For these reasons, the main source of data for vaccine outcomes was the DSHS school-vaccination coverage data which provides the proportion of kindergarten students in public and private schools that meet the measles immunization requirement of two doses of the measles, mumps and rubella (MMR) vaccine and six other antigens.

County Health Rankings (CHR) data which I augmented with five additional county-level data including 1.) County Health Rankings data which collates various health and socio-demographic indicators from multiple Texas state agencies, 2.) data on Health Professional Shortage Area (HPSA) scores, 3.) data on Primary care physicians scores; 4.) data on Medically Underserved Area (MUA) scores accessed via BigQuery, as well as 5.) data with variables related to COVID-19 vaccine hesitancy collected via the Household Pulse Survey.

Annex 1 contains a full list of variables and the source of each variable. Data was available from all sources up and including 2023, so this is the most recent data included in the analysis. An exception was the MUA, HPSA, and Primary care physician county-level scores. These files contained data that was not regularly updated for certain counties, so the most recent data available at the time of analysis for each county was used.

It is also worth noting that one research study used both forward and backward step-wise selection of 105 potential covariates in order to estimate county-level coverage rates (Seeskin et al. 2025). Three of the covariates used in that study's analysis were similar to those included in this analysis such as the children's health insurance status, food environment index, limited access to healthy foods and one covariate was the same – the percentage of residential segregation (black/white). In addition, Seeskin and colleagues also selected indicators such as the number of hepatitis A vaccine doses administered relative to the population of children, the percentage of the civilian labor force employed in education/health care/social assistance, and the percentage of households with a female head.

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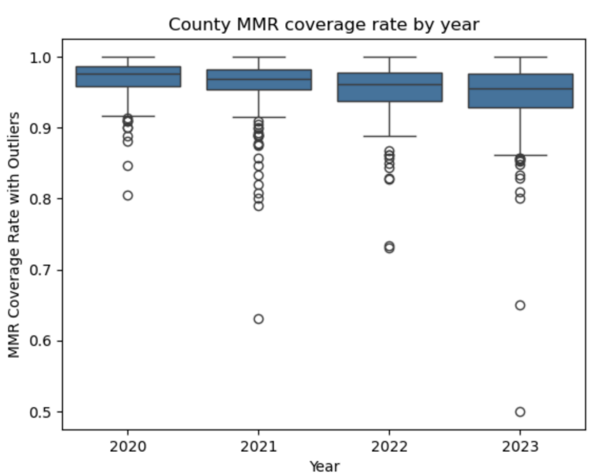
Collation of Data Sets

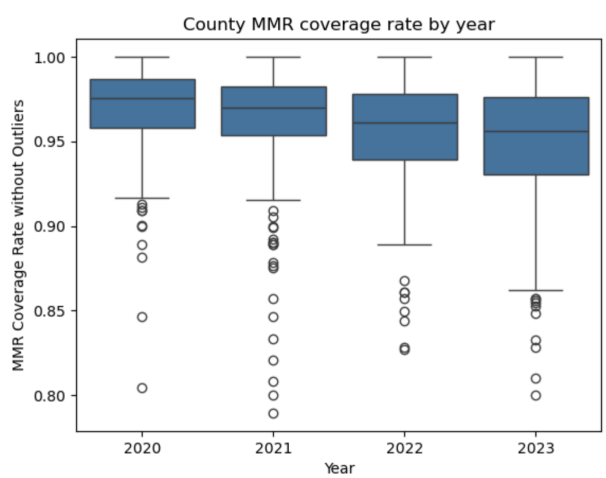
All data was downloaded in csv files from the respective websites. The names of 254 counties were used as keys to collate and combine the data sets for each of the four years of data across the five data sources. Data was checked for duplicates and missing values before and after merging such that each row represented all the information possible for a given county and year of data.

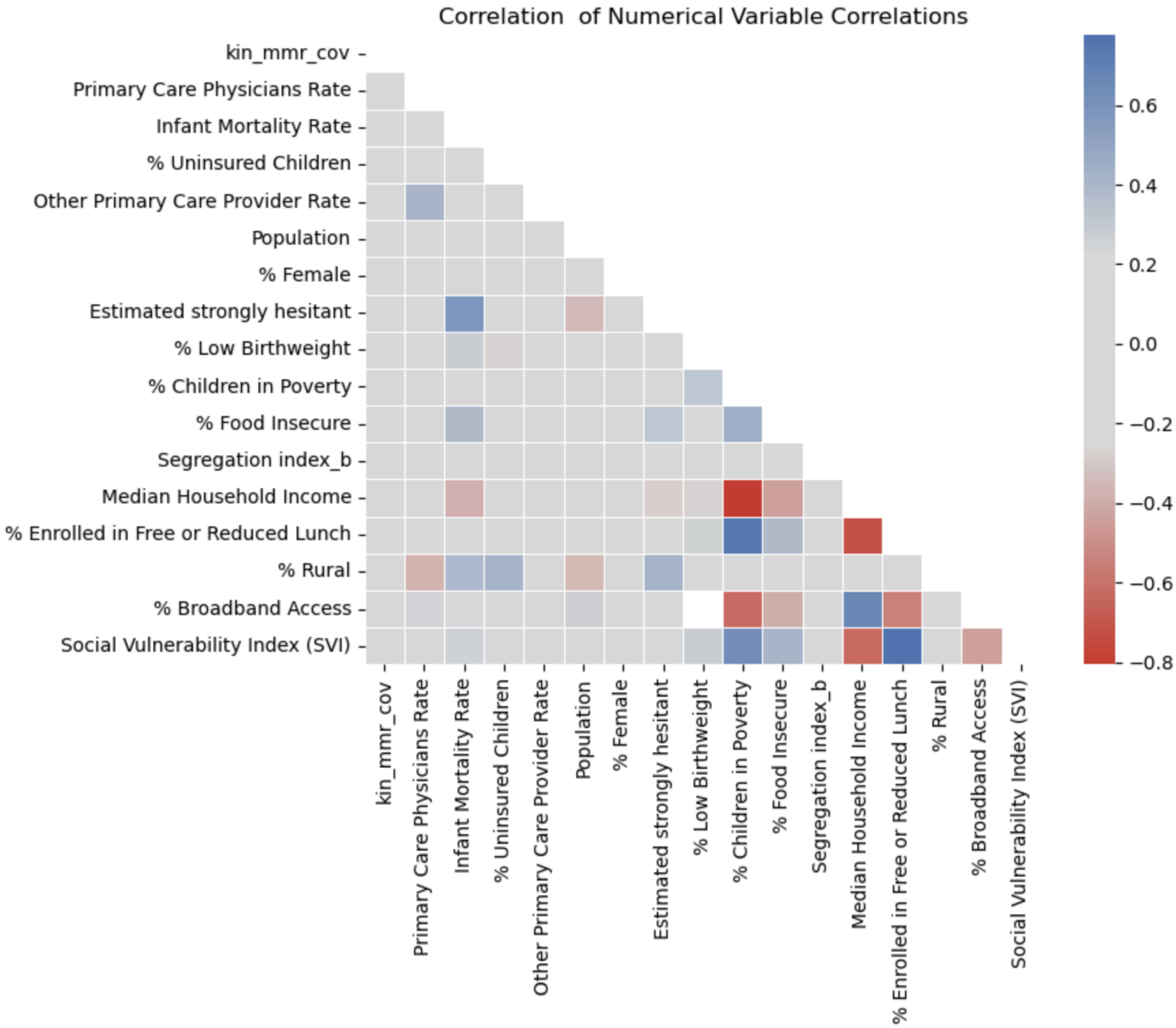
Using available documentation, I selected and used the HPSA score 'Geographic' type as this reflects a shortage of providers for an entire group of people within a defined geographic area, which was likely to best correspond to my county-level analysis and where there were still multiple rows of data for a single year and county, these were replaced by the average HPSA scores for that year and county. A similar approach was used for MUA scores and the provider per 1000 people ratio in the MUA data set.

Exploratory data Analysis

After merging and combining the various data sets, exploratory analysis was conducted on 34 numeric variables by visualizing the distribution of each variable and its association with other variables. The distribution of many variables such as county population, the proportion of a county population with broadband internet access, and all vaccination coverage rates were significantly skewed. Given the ceiling effect observed in MMR vaccination rates, five county-year outlying values with MMR coverage values equal or less than 79% (1% percentile) were dropped.







The heat map above shows strong associations (collinearity) among several features including median household income, the proportion of children in poverty, the proportion enrolled in free or reduced lunch, and the proportion of residents with broadband internet access.

Eight variables were missing 25% or more of their values. Seven counties had missing MMR coverage rate values for at least one year and three counties – DeWitt, Loving, and McMullen were missing data for all four years so these counties were dropped from the analysis and modeling in further analysis. Three approaches were used to impute missing values. First, values for a given variable were imputed based on whether the magnitude of percent change for a given county's value was equal or greater to 3% from 2020 to 2023. For example, variables like the percentage of females in a county or the percentage of the population living in rural settings did not change from year to year. Because they changed less than 3% from 2020 to 2023, missing values for each county were replaced with the median of non-missing values for that county. Second, linear interpolation was used to impute county-specific values between, after, and before non-missing values. Third, global imputation was used to impute the remaining missing values for county-years that could not be interpolated.

Model Construction and Scenario modelling

To examine the features most important in explaining and predicting MMR coverage in new counties, a group split was performed. To test the ability of models to predict future (2023) MMR coverage rates based on the first three years of training data for each county, a temporal train/test split was performed. For both splits, cross validation on five folds of shuffled training data and SelectKBest was used to limit the number of features.

Most important predictors of coverage

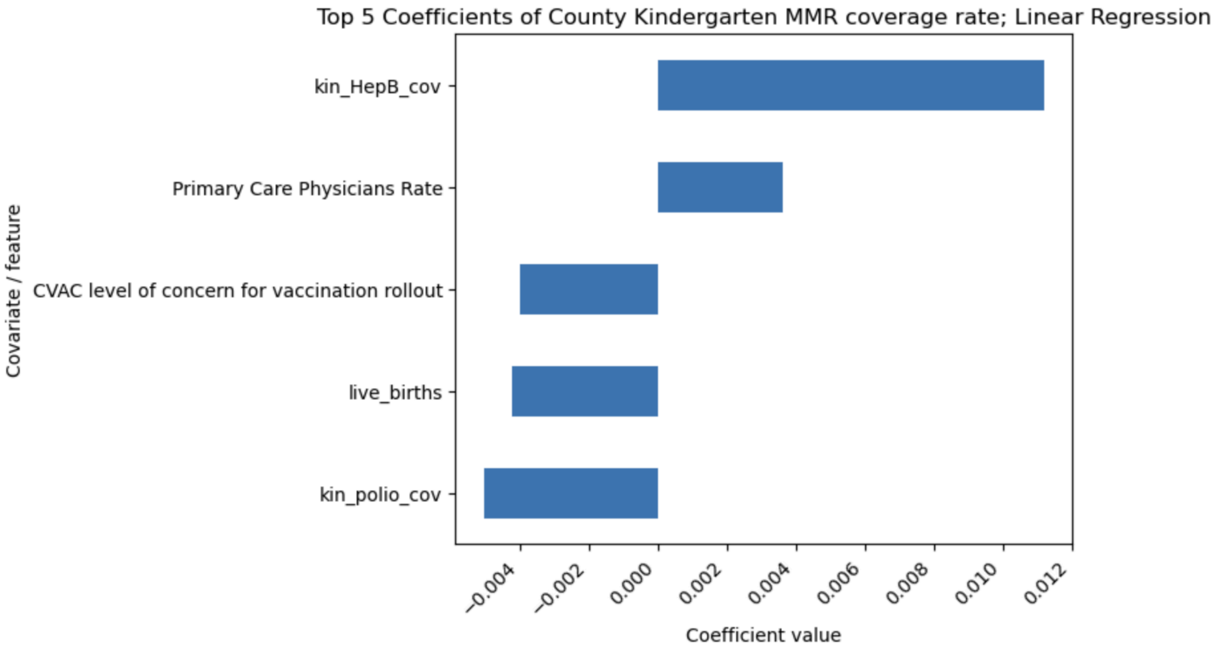
For the group split, a 75/25 train test split was performed by county such that data for each county was split into either the training or test data sets. This was done so that models could not use information on a particular county to predict MMR coverage during testing. Table 1 shows results of those models trained and tested on the data split by group. The XGboost did not identify any features as being important so that model was not used to predict coverage on the test data. Negative R squared values demonstrate that most models failed to predict coverage based on the covariates or features in the data.

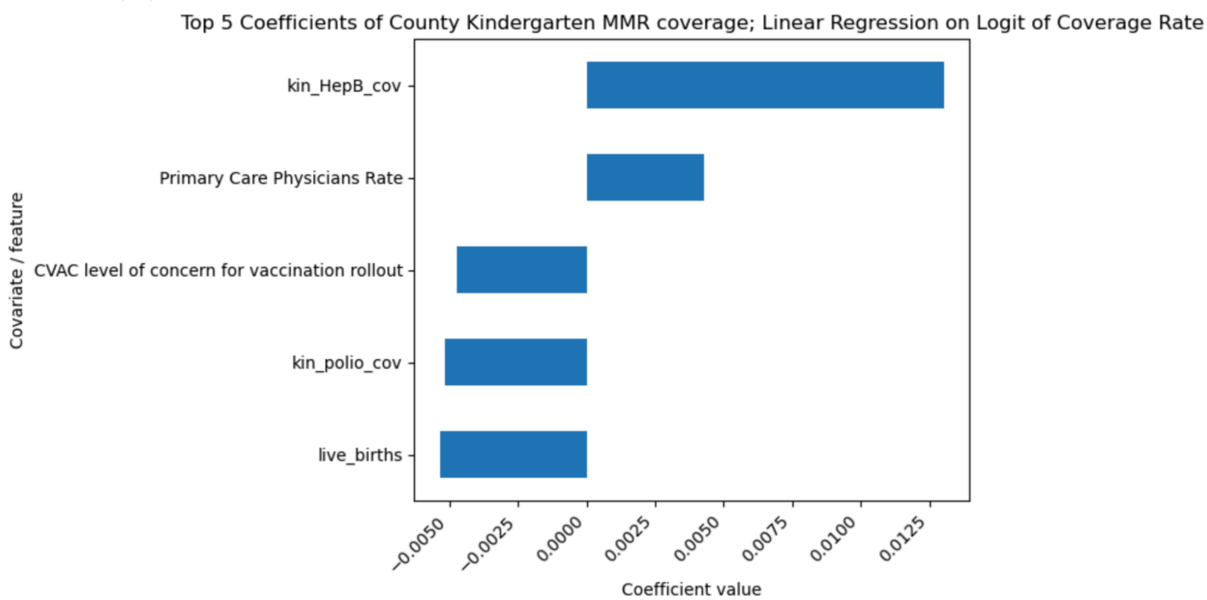
Table 1: Model metrics for estimating new county MMR coverage rates

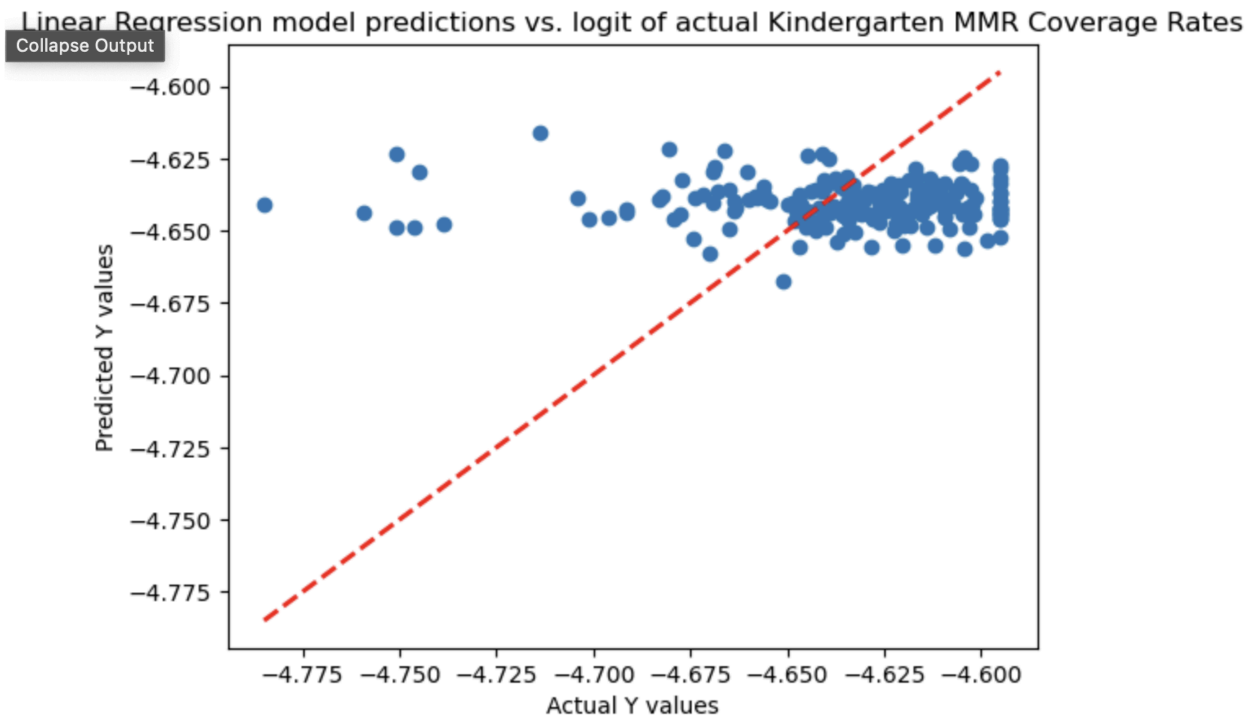
Model	Training data		Test data	
	R squared	RMSE	R squared	RMSE
Linear Regression	0.025	0.0434	-0.0968	0.0320
Ordinary Least Squares	0.034	0.0425	0.0849	0.0310
Ridge	-0.0108	0.0430	-0.0553	0.0010
Random Forest	0.0584	-0.00174	0.0584	-0.00105
XGBoost	-0.00000	-0.00185	N/A	N/A

The best resulting model was the linear regression model based on the highest RMSE on training data and test data. The distribution of coverage was highly skewed to the left with 70% and 77% of values in the training and testing data split by county above 95%, respectively. Due to these severe ceiling effects, a logit transformation was applied to the percentage of children vaccinated (p) whereby $logit(p)=log(1-pp)$ in order to expand differences in values close to 100% coverage. Unfortunately this did not result in an improved model performance.

The strongest predictors of coverage differed slightly based on whether or not a logit transformation of coverage was applied. In either case, the model fit remained low (RMSE 0.008 on training, and 0.0356 on test when coverage was transformed to its log odds)







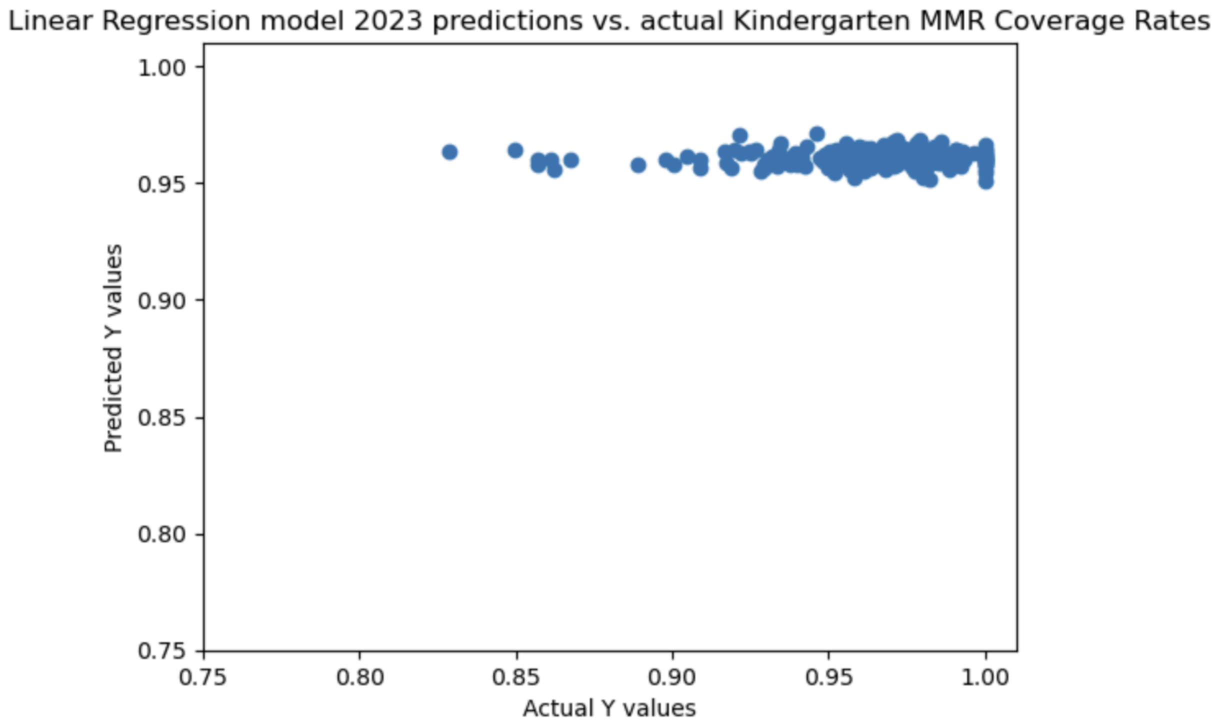
Predicting 2023 coverage

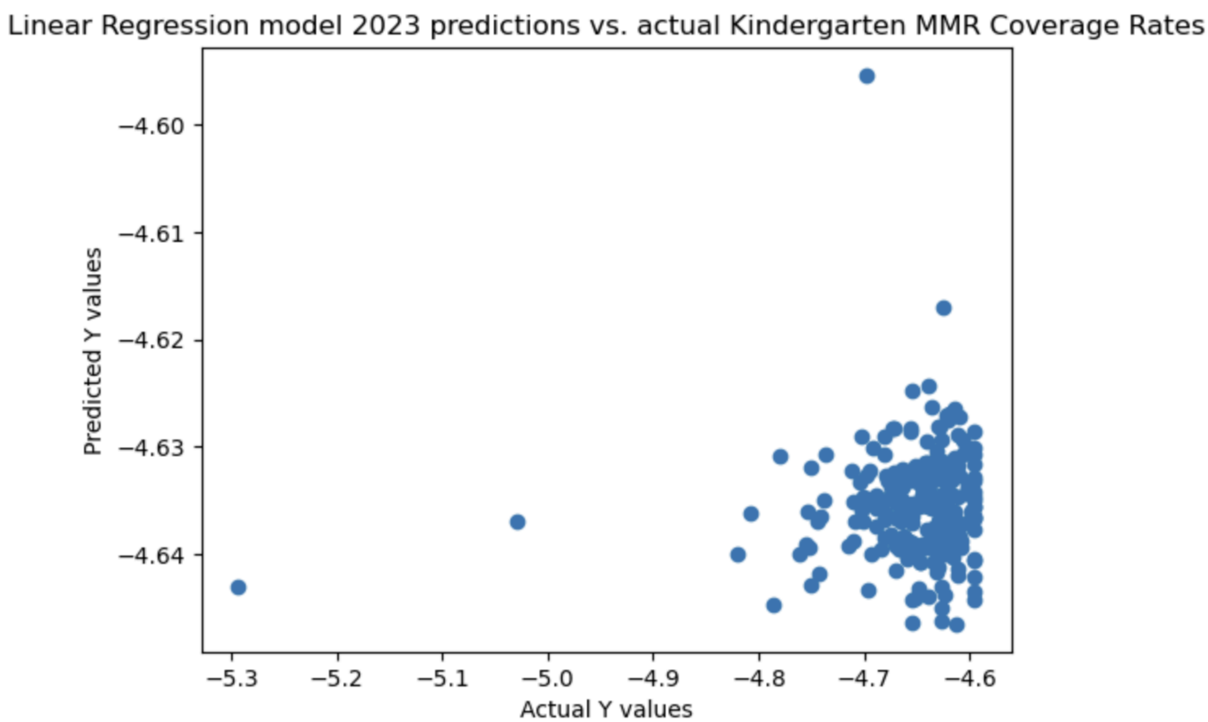
The temporal split involved dedicating three years of data (2020-2022) for all counties to training and the 2023 data from all counties for model testing. The performance of the linear regression model on training and test data was still dismally low with or without the logit coverage transformation.

Table 2: Model metrics for predicting 2023 county MMR coverage rates

Model		**Training data**		**Test data**	
		R squared	**RMSE**	**R squared**	**RMSE**
Coverage	Linear Regression	0.0125	0.0362	-0.00786	0.0316
Log Odds of Coverage		0.0126	0.0405	-0.0701	0.0648

Predicting log odds of MMR coverage instead of MMR coverage slightly improved model performance but it was still very low. When plotting predicted 2023 coverage y vs actual MMR coverage, we see that transformation of the outcome may improve prediction but not drastically so.





The most important features differed based on whether coverage was transformed to log odds of coverage rate. When predicting MMR coverage in 2023, the linear regression model found the same five features to be most important as when estimating coverage for new counties after the group split. However, instead of Hepatitis B, CVAC Index and the Primary Care Physicians rate, the model predicting 2023 log transformed values found the Black/White segregation index, infant mortality rate, and the # of primary care physicians to play an important role.

Conclusions

1. **What are the strongest predictors of high county-level measles vaccination rates across Texas being up-to-date on the MMR vaccine series?**

The strongest five predictors of kindergarten MMR coverage at the county level in Texas based on the socio-demographic features that could be gathered from publicly available data were 1.) Hepatitis B coverage, 2.) the number of live births in a county, 3.) polio vaccine coverage, 4.) the concern for COVID-19 vaccination rollout index and 5.) the Primary Care Physicians Rate. These predictors were also the most important when predicting the log odds of MMR coverage with almost no differences in the direction or magnitude of model coefficients based on where the MMR coverage or the log odds of MMR coverage were used.

- Counties with a larger number of live births were less likely to obtain higher coverage rates, perhaps due to limitations in available resources to serve a fast growing population.
- The COVID-19 vaccination rollout index (CVAC) reflects a county's ability to handle a COVID-19 outbreak and is based on a community's access to health care, affordable housing, transportation, childcare, or safe and secure employment. CVAC Index values range from 0 (least vulnerable) to 1 (most vulnerable); so the linear regression model suggests the higher the less ready a county was to handle COVID-19 during the pandemic, the more likely they are to have a lower MMR coverage rate today.
- It is interesting to note that while Hep B coverage rates are positively correlated with MMR coverage whereas polio coverage was inversely correlated.
- The Primary Care Physicians Rate, or the number of primary care physicians per 100,000 people, was positively correlated with MMR coverage among kindergarteners. This suggests that MMR coverage is influenced by adequate access to primary care physicians.
- However, the coefficients on these models were extremely low and model performance was poor. Further model optimization may be needed to identify a better performing model and obtain more precise coefficients estimates.

2. **What ability do machine learning models have in predicting future (i.e. 2023) county-level measles vaccination immunization coverage rates based on data from previous years?**

Among the various models tested, the linear regression model appeared to best predict 2023 kindergarten MMR coverage in Texas based on the available data from 2020 to 2022. Future MMR coverage was most influenced by 1.) polio coverage, 2.) the number of live births in a

county, 3.) the infant mortality rate, 4.) the number of primary care physicians and 5.) the Black/White segregation index.
However, the performance of the linear regression model was extremely low and warrants further optimization if it is to be useful.

The West-Texas measles outbreak was a reminder that school-age vaccination coverage has direct consequences on the transmission of vaccine-preventable diseases. The development of such models to explain or predict future coverage rates have potential to empower state-level public health officials to leverage the drivers and barriers of immunizations uptake among children entering the school system. Information on coefficients, for example, shed light on potential strategies to improve MMR coverage rates such as supporting programs designed to increase the number of primary care physicians in a county, prioritize funding for counties with the greatest number of live births and infant mortality rates.

Annex 1:

Type of county-level covariate	Covariate/Feature	Data Source
Health care access	Percentage of uninsured children*	countyhealthrankings.org
	Health Professional Shortage Area (HPSA) - Primary care physicians score; Medically Underserved Area (MUA) score - Primary care physicians	Health Resources & Services Administration (HRSA) via BigQuery
Economic	Percentage of population that is food insecure*	countyhealthrankings.org
	Median household income	countyhealthrankings.org
	Percentage of children in poverty	countyhealthrankings.org
Demographic	Residential segregation (black/white)*	countyhealthrankings.org CDC
	Percentage of population that is children (0-18 years)	countyhealthrankings.org
	Percentage of residents in living in rural areas within the county	countyhealthrankings.org
	# of live births in each county (2019-2022)	https://healthdata.dshs.texas.gov/dashboard/births-and-deaths/live-births
Individual Psychosocial	COVID-19 vaccine hesitancy (from June 2021) Social vulnerability index (SVI) COVID-19 Vaccine Coverage Index (CVAC)	CDC
Other health outcomes	Infant mortality	countyhealthrankings.org
	Number of flu vaccinations administered	countyhealthrankings.org

	Number of infants with low birthweight	countyhealthrankings.org